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in ana-correia

🐙 raquelvcorreia

PROFILE

- Experienced wet-lab **protein biochemist**, evolved into a **computational designer**.
- Passionate about solving complex **biological puzzles** by combining structural and experimental insights with **AI/ML pipelines** to design the next generation of functional proteins.

KEY SKILLS

- Computational Protein Design and AI/ML.
- Structural Biology & Protein Characterization.
- Data Science, Analytics & Infrastructure.
- R&D Project Management & Collaboration.
- Python, MongoDB, SQL, R, AWS, and basic Apache Spark for ML.

ANA R. CORREIA

Work Experience

SENIOR AI (PROTEIN) ENGINEER | SAMSARA ECO NSW, AUSTRALIA | Jul 2024 - Present

- Protein design and engineering of novel enzymes for plastic recycling applications.
- Benchmarking and optimization of in-house ML models, while proactively auditing and integrating cutting-edge, open-source AI/ML protein engineering tools into discovery platforms.
- Integrate ML predictions with structural, biophysical, and experimental data to accurately prioritize high-quality variants for validation.
- Collaborate on architecting the company's data model and building the database infrastructure, to enhance data accessibility between teams and improve integration across engineering pipelines.
- Bridge computational and wet-lab teams to drive rapid design-build-test-learn iterations for novel enzyme development.

SCIENTIST | AMGEN INC. Thousand Oaks, CA, USA | Apr 2018 - Sept 2022

- Integrated multidisciplinary teams focused on leveraging and structuring data for ML/AI applications.
- Project development and management.
- Protein engineering, development of purification and analytical characterization workflows for established and novel antibody-like formats (including bi- and multivalent biologics).
- Mentoring, cross-training and supervision of junior scientists in purification, data analysis, and high throughput sample tracking systems.

POSTDOC | CALTECH | Hoelz Lab Pasadena, CA, USA | Nov 2013 - Mar 2018

- Expression, purification, assembly and characterization of large macromolecular complexes.
- Structural determination of protein complex by X-ray crystallography.
- Expression and characterization of synthetic antibodies, for further use in crystallization assays.

POSTDOC | CNB-CSIC | Caston Lab Madrid, SPAIN | Aug 2011 - Oct 2013

- Production of virus in mammalian and fish cell lines. Purification of virus and viral like particles and subsequent biochemical and structural characterization (negative-stain and Cryo-EM).
- Viral ribonucleoprotein complexes extraction and characterization.

VISITING RESEARCHER | BONN UNIVERSITY | Voos Lab Bonn, GERMANY | Feb - Jul 2010

- In vivo (yeast cells) and ex vivo (isolated mitochondria) characterization of a mitochondrial protein biogenesis.

Education

PHD STUDENT | ITQB-NOVA | Gomes Lab Lisbon, PORTUGAL | 2006 - 2010

- In vivo (yeast cells) and ex vivo (isolated mitochondria) characterization of a mitochondrial protein biogenesis.

INTERN AND VISITING PHD STUDENT | NIMR-MRC London, UK | 2005 - 2009

- In vivo (yeast cells) and ex vivo (isolated mitochondria) characterization of a mitochondrial protein biogenesis.

BACHELOR'S DEGREE, BIOCHEMISTRY, 2005 | UNIVERSITY OF LISBON (FCUL)

LANGUAGES

- English
- Spanish
- Portuguese (native)

HOBBIES & INTERESTS

- Hiking
- Scuba Diving
- Ocean Conservancy
- Mountain Biking
- Traveling

Selected Additional Training & Certifications

- **MicroMasters Programs MITx – Statistics and Data Science (2025).**
- Python 3 Programming by University of Michigan on Coursera (2022).
- Applied Machine Learning in Python by University of Michigan on Coursera (2023).
- Machine Learning Specialization by Stanford University & DeepLearning.AI on Coursera (2023).
- Deep Learning Specialization by DeepLearning.AI on Coursera (2023).
- Data Science Specialization by Johns Hopkins University on Coursera (8 courses from the specialization including R programming, 2021).
- Bioinformatics I, II by University of Toronto on Coursera (2023).
- Algorithms for DNA sequencing by Johns Hopkins University on Coursera (2023).
- SQL for Data Science by University of California, Davis on Coursera (2023).
- Apache Spark for Data Engineering and Machine Learning by IBM on edX (2023)

Publications (8 as a first author, *)

- Utility of physiologically based pharmacokinetic modeling to predict inter-antibody variability in monoclonal antibody pharmacokinetics in mice. (2023) **mAbs**, 15(1): 2263926.
- Development of in silico models to predict viscosity and mouse clearance using a comprehensive analytical data set collected on 83 scaffold-consistent monoclonal antibodies (2023) **mAbs**, 15(1):2256745.
- Architecture of the cytoplasmic face of the nuclear pore (2022) **Science**, 376 (6598):1174-1192.
- Enhancing the Prefusion Conformational Stability of SARS-CoV-2 Spike Protein Through Structure-Guided Design (2021) **Front. Immunol.**, 12, article 660198.
- Rational selection of building blocks for the assembly of bispecific antibodies (2020) **mAbs**, 13 (1), e1870058.
- Structural and functional analysis of mRNA export regulation by the nuclear pore complex (2018) **Nat. Commun.**, 9 (1):2319-2338. *
- A new MR-SAD algorithm for the automatic building of protein models from low-resolution X-ray data and a poor starting model (2018) **IUCr**, 5 (Prt 2):166-171.
- Nuclear pores. Architecture of the nuclear pore complex coat (2015) **Science**, 347, 1148-52. *
- Probing the kinetic stabilities of Friedreich's Ataxia clinical variants using a solid phase GroEL chaperonin capture platform (2014) **Biomolecules**, 4 (4): 956-79. *
- Revertants, low temperature, and correctors reveal the mechanism of F508del-CTFR rescue by VX-809 and suggest multiple agents for full correction (2013) **Chem. Biol.**, 20 (7):943-55.
- Iron binding activity in yeast frataxin entails a trade off with stability in the a1/b1 acidic ridge region (2010) **Biochem. J.**, 426(2):197-203. *
- No evidence of direct binding between ursodeoxycholic acid and the p53 DNA-binding domain (2010) **Biosci Rep.**, 30(5):359-64. *
- The conserved Trp-155 in human frataxin as a hotspot for oxidative stress related chemical modifications (2009) **BBRC**, 390(3):1007-11. *
- Dynamics, stability and iron-binding activity of frataxin clinical mutants (2008) **FEBS J.**, 275, 3680-3690. *
- Conformational stability of human frataxin and effect of Friedreich's ataxia related mutations on Protein Folding (2006) **Biochem. J.**, 398, 605-611. *